

GAUSSIAN SMOOTHING AND THE DIMENSION OF SELF-SIMILAR MEASURES ON THE LINE

GPT-6 ASTRA

ABSTRACT. We present a Gaussian-channel approach to the dimension theorem for self-similar measures on the real line. The argument allows unequal contraction ratios, reflections, and exact overlaps, and uses the random-walk entropy of the law on similarities. Its main ingredient is an elementary entropy-growth estimate: a lower bound on the derivative of a Gaussian-smoothed measure makes the successive digits of an independent perturbation detectable. The information chain rule adds these detections without any independence assumption on the digits. Self-similarity supplies the derivative bound and an exact identity for the information carried by a composed map. Together these yield superexponential concentration whenever the dimension is smaller than both one and the ratio of random-walk entropy to the Lyapunov exponent.

This is an unedited model output that led to the paper [KK26]. Prompting was done by C.K. and the prompter's local context was used.

1. INTRODUCTION

Let μ be a finitely supported probability measure on the similarities of $\mathbb{R}$, written as

$$\mu = \sum_{i \in \Lambda} p_i \delta_{\varphi_i}, \quad \varphi_i(x) = a_i x + b_i, \quad 0 < |a_i| < 1.$$

The maps φ_i are distinct, $p_i > 0$, and $\sum_i p_i = 1$. Write $r_i = |a_i|$. The stationary measure ν is the unique probability measure satisfying

$$\nu = \mu * \nu = \sum_i p_i (\varphi_i)_* \nu. \tag{1.1}$$

Existence and compact support follow from Hutchinson [Hut81], and exact dimensionality from Feng–Hu [FH09]; we denote its dimension by α .

We use the notation of Kittle–Kogler [KK25]. Thus

$$\chi_\mu = \sum_i p_i \log r_i < 0, \quad \chi = |\chi_\mu|, \quad h_\mu = \lim_{n \rightarrow \infty} \frac{H(\mu^{*n})}{n}. \tag{1.2}$$

Here H is Shannon entropy, with natural logarithms. The last limit exists by subadditivity. If two words give the same similarity, their probabilities are added in μ^{*n} . Consequently, h_μ counts distinct maps with their induced probabilities; it need not equal the one-step entropy $-\sum_i p_i \log p_i$.

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This manuscript presents a proposed proof which has not yet been independently verified.

For a similarity $g(x) = a(g)x + b(g)$, put $r(g) = |a(g)|$ and $\sigma(g) = \text{sgn } a(g)$. Equip the similarity group with the metric

$$d(g, h) = |\log r(g) - \log r(h)| + |\sigma(g) - \sigma(h)| + |b(g) - b(h)|,$$

and define

$$\Delta_n = \min\{d(g, h) : g, h \in \text{supp}(\mu^{*n}), g \neq h\}. \quad (1.3)$$

The minimum of an empty set is $+\infty$. Since this minimum is over distinct maps, an exact overlap does not make Δ_n zero.

Theorem 1.1. *Let μ and ν be as above. If*

$$\dim \nu < \min \left\{ 1, \frac{h_\mu}{|\chi_\mu|} \right\},$$

then

$$\lim_{n \rightarrow \infty} -\frac{1}{n} \log \Delta_n = +\infty. \quad (1.4)$$

*In fact, for every $C > 0$ and every sufficiently large n , there are distinct $g, h \in \text{supp}(\mu^{*n})$ with*

$$a(g) = a(h), \quad |b(g) - b(h)| < e^{-Cn}.$$

In particular, if $\Delta_n \geq e^{-cn}$ for some $c > 0$ and infinitely many n , then

$$\dim \nu = \min \left\{ 1, \frac{h_\mu}{|\chi_\mu|} \right\}. \quad (1.5)$$

Hochman [Hoc14] established the dimension theorem on the line using an inverse theorem for entropy. Kittle–Kogler [KK25] developed a proof by variance summation, allowing exact overlaps and, more generally, contraction on average. The variance-summation method originates in Kittle [Kit25]. Their separation quantity is

$$M_n = \min \left\{ d(g, h) : g, h \in \bigcup_{j=0}^n \text{supp}(\mu^{*j}), g \neq h \right\}.$$

Since $M_n \leq \Delta_n$, their exponential separation hypothesis along infinitely many levels implies the hypothesis in Theorem 1.1. Throughout this paper every generator is strictly contracting.

The argument begins with the identity $I(A; X) = \alpha \chi$ for $X \sim \nu$ and its first symbol A . When $\alpha < 1$, it gives a uniform lower bound $t \|f'_t\|_1 \geq \kappa > 0$ for the density f_t of $X + tZ$, where Z is an independent standard Gaussian. Our main analytic observation is that this bound forces entropy growth: for every independent $Y \in [0, 1)$, every $\eta > 0$, and $0 < t \leq 1$,

$$H(\lfloor Y/t \rfloor) \leq \eta \log(1/t) + C_\eta (1 + I(Y; X + Y + tZ)). \quad (1.6)$$

We prove this by testing successive digits of Y at separate Gaussian noise levels and adding their contributions through the information chain rule. A composed map, conditioned on its location at its natural scale, carries only $o(n)$ further information about the final point. Exponential separation would give its remaining translation linear entropy, in contradiction with (1.6).

2. SELF-SIMILARITY AND GAUSSIAN SMOOTHING

We write h for differential entropy and I for mutual information. All Gaussian variables introduced below are standard and independent of the other variables in the same expression, unless stated otherwise. For a positive smooth density f , its Fisher information is

$$J(f) = \int_{\mathbb{R}} \frac{|f'(x)|^2}{f(x)} dx.$$

We first recall how Gaussian smoothing compares with quantization.

Lemma 2.1. *For a bounded real random variable X and $t > 0$,*

$$h(X + tZ) = H(\lfloor X/t \rfloor) + \log t + O(1),$$

where the constant is absolute.

Proof. Write $X/t = K + U$, where $K \in \mathbb{Z}$ and $U \in [0, 1)$. The chain rule gives

$$I(X; X + tZ) = I(K; K + U + Z) + I(U; U + Z | K).$$

The second term is bounded by the Gaussian maximum-entropy inequality. Also $K - \lfloor K + U + Z \rfloor = -\lfloor U + Z \rfloor$ has uniformly bounded entropy, so $H(K | K + U + Z) = O(1)$. Thus the displayed expression equals $H(K) + O(1)$. Now use $h(X + tZ) = I(X; X + tZ) + h(tZ)$. $\square$

Take $X \sim \nu$, and let f_t denote the density of $X + tZ$. Exact dimensionality and compact support give

$$H(\lfloor X/t \rfloor) = \alpha \log(1/t) + o(\log(1/t)) \quad (t \downarrow 0). \quad (2.1)$$

In particular, if $h_t = h(f_t)$, then

$$h_{e^{-s}} = -(1 - \alpha)s + o(s) \quad (s \rightarrow \infty). \quad (2.2)$$

Couple $X = \varphi_A(X')$, where $\mathbb{P}(A = i) = p_i$ and X' is an independent copy of X .

The first-symbol identity below is the Bernoulli similarity case of Feng–Hu’s projection-entropy formula [FH09, Theorem 2.8]. We give a direct proof in the present notation.

Proposition 2.2. *The first-symbol information satisfies*

$$I(A; X) = \alpha \chi. \quad (2.3)$$

If $\alpha < 1$, then $t \|f'_t\|_1 \geq \kappa$ for all $t > 0$, with

$$\kappa = \frac{c_0^{3/2}}{\sqrt{3}}, \quad c_0 = \frac{2(1 - \alpha)\chi}{\sum_i p_i(r_i^{-2} - 1)}. \quad (2.4)$$

Proof. Set $I_t = I(A; X + tZ)$. Conditioning on A gives the exact identity

$$I_t = \chi + h_t - \sum_i p_i h_{t/r_i}. \quad (2.5)$$

The quantity I_t increases as $t \downarrow 0$ by data processing, and is bounded by $H(A)$. To identify its limit, put $\ell_i = \log(1/r_i)$ and average over $t = e^{-s}$, $0 \leq s \leq T$. For each fixed $\ell > 0$, (2.2) gives

$$\begin{aligned} \frac{1}{T} \int_0^T (h_{e^{-s}} - h_{e^{-(s-\ell)}}) ds &= \frac{1}{T} \left(\int_{T-\ell}^T h_{e^{-s}} ds - \int_{-\ell}^0 h_{e^{-s}} ds \right) \\ &\longrightarrow -(1 - \alpha)\ell. \end{aligned}$$

Since $\sum_i p_i \ell_i = \chi$, the limit of I_t is $\alpha\chi$. It equals $I(A; X)$: data processing gives one inequality, and lower semicontinuity of relative entropy under the weak convergence $(A, X + tZ) \rightarrow (A, X)$ gives the other. This proves (2.3) and also $I_t \leq \alpha\chi$ for every $t > 0$.

De Bruijn's identity [Sta59] is $\frac{d}{dt} h_t = tJ(f_t)$. Fisher information decreases under Gaussian convolution. Hence

$$\begin{aligned} (1 - \alpha)\chi &\leq \chi - I_t = \sum_i p_i \int_t^{t/r_i} uJ(f_u) du \\ &\leq \frac{t^2 J(f_t)}{2} \sum_i p_i (r_i^{-2} - 1). \end{aligned} \quad (2.6)$$

In particular, $t^2 J(f_t) \geq c_0$.

Let $S_t = f'_t/f_t$. The Gaussian score identity reads

$$S_t(X + tZ) = -t^{-1} \mathbb{E}[Z \mid X + tZ].$$

It implies $\mathbb{E}|S_t(X + tZ)|^4 \leq 3t^{-4}$. Interpolation between L^1 , L^2 , and L^4 therefore gives

$$\|f'_t\|_1 = \mathbb{E}|S_t(X + tZ)| \geq \frac{J(f_t)^{3/2}}{(\mathbb{E}|S_t(X + tZ)|^4)^{1/2}} \geq \frac{c_0^{3/2}}{\sqrt{3}t}.$$

This proves the derivative bound. $\square$

3. AN ENTROPY ESTIMATE FOR ADDITIVE PERTURBATIONS

In this section X is any bounded real random variable whose smoothed densities satisfy $t\|f'_t\|_1 \geq \kappa > 0$ for all $t > 0$. We may decrease κ so that $\kappa \leq 1$. No self-similarity assumption is used here.

Proposition 3.1. *For every $\eta > 0$ there is $C_\eta < \infty$, depending only on η and κ , such that every independent $Y \in [0, 1)$ and every $0 < t \leq 1$ satisfy*

$$H(\lfloor Y/t \rfloor) \leq \eta \log(1/t) + C_\eta (1 + I(Y; X + Y + tZ)). \quad (3.1)$$

The same constants apply to $-X$. Moreover,

$$I(Y; X + Y + tZ) = h(X + Y + tZ) - h(X + tZ).$$

The proof rests on two elementary observations. A small translation is detected to first order through its conditional mean, and the entropy of a digit is controlled by the variance of those means.

Lemma 3.2. *Let $U \in [0, a]$ and a finite random variable L be jointly distributed and independent of X, Z . Then*

$$I(L; X + U + tZ) \geq \frac{1}{4} \|f'_t\|_1^2 \text{Var}(\mathbb{E}[U \mid L]) - \frac{2a^4}{t^4}.$$

Proof. The Gaussian derivative bound $\|f''_t\|_1 \leq 2t^{-2}$ implies

$$\|f_t(\cdot - u) - f_t + uf'_t\|_1 \leq a^2/t^2 \quad (0 \leq u \leq a).$$

Let g_l and g be the conditional and unconditional output densities. Then

$$g_l - g = -(\mathbb{E}[U \mid L = l] - \mathbb{E}U)f'_t + R_l, \quad \|R_l\|_1 \leq 2a^2/t^2.$$

Pinsker's inequality (see [CT06, Lemma 11.6.1]) $D(g_l \| g) \geq \frac{1}{2} \|g_l - g\|_1^2$, together with $(x - y)_+^2 \geq x^2/2 - y^2$, yields the result after averaging over l . $\square$

Lemma 3.3. *Let $U \in [0, 1)$, let $M \geq 2$ be an integer, and put $Q = \lfloor MU \rfloor$ and $v = \text{Var}(\mathbb{E}[U \mid Q])$. Then*

$$H(Q) \leq \frac{1}{2} \log(4\pi e(1 + M^2 v)).$$

Proof. Since $Q - M\mathbb{E}[U \mid Q] \in [-1, 0]$, we have $\text{Var } Q \leq 2M^2 v + 1/2$. For an independent uniform random variable V on $[0, 1)$, the Gaussian maximum-entropy inequality gives

$$H(Q) = h(Q + V) \leq \frac{1}{2} \log(2\pi e(\text{Var } Q + 1/2)).$$

The claimed bound follows. $\square$

Proof of Proposition 3.1. Fix an integer $M \geq 2$, to be chosen later. For $0 \leq j \leq k$, set

$$a_j = M^{-j}, \quad K_j = \lfloor Y/a_j \rfloor, \quad G_k = I(Y; X + Y + a_k Z),$$

and for $0 \leq j < k$ put

$$v_j = a_j^{-2} \mathbb{E}[\text{Var}(\mathbb{E}[Y \mid K_{j+1}] \mid K_j)].$$

The variables K_j reveal successive base- M digits of Y , with $K_0 = 0$. For $A \geq 1$, the information chain rule and data processing give

$$\begin{aligned} G_k &\geq \sum_{j=0}^{k-1} I(K_{j+1}; X + Y + a_k Z \mid K_j) \\ &\geq \sum_{j=0}^{k-1} I(K_{j+1}; X + Y + Aa_j Z \mid K_j) \\ &\geq \frac{\kappa^2}{4A^2} \sum_{j=0}^{k-1} v_j - \frac{2k}{A^4}. \end{aligned} \tag{3.2}$$

For the second line, add Gaussian noise separately in each summand. For the third, condition on K_j and subtract the left endpoint of its interval. The remaining translation lies in $[0, a_j)$, so Lemma 3.2 applies at noise level Aa_j . Conditioning on K_j leaves the law of X unchanged. The digits of Y need not be independent.

Apply Lemma 3.3 in each parent interval, and use Jensen's inequality first over intervals and then over j . Writing $\bar{v} = k^{-1} \sum_{j < k} v_j$, we obtain

$$H(K_k) = \sum_{j < k} H(K_{j+1} \mid K_j) \leq \frac{k}{2} \log(4\pi e(1 + M^2 \bar{v})). \tag{3.3}$$

Choose $A^2 = 8M^2/\kappa^2$. Inequality (3.2) gives

$$\bar{v} \leq \frac{4A^2}{\kappa^2 k} G_k + M^{-2}.$$

Substitute this in (3.3) and use $\log(1+x) \leq x$. It follows that

$$H(K_k) \leq \frac{k}{2} \log(8\pi e) + \frac{8M^4}{\kappa^4} G_k. \tag{3.4}$$

For $k = 0$ the same inequality holds trivially.

Now choose M so large that $\log(8\pi e)/(2\log M) \leq \eta$. Given $0 < t \leq 1$, let $k = \lfloor \log(1/t)/\log M \rfloor$, so $t \leq a_k < Mt$. Each a_k -interval meets at most $M + 1$ t -intervals. Consequently,

$$H(\lfloor Y/t \rfloor) \leq H(K_k) + \log(M + 1).$$

Also $G_k \leq I(Y; X + Y + tZ)$ by data processing. Equation (3.4) proves (3.1). Reflection preserves the derivative bound, and the final identity follows by conditioning on Y . $\square$

4. INFORMATION CARRIED BY A COMPOSED MAP

Let $I_1, I_2, \dots$ be independent with $\mathbb{P}(I_j = i) = p_i$, and write

$$W_n = (I_1, \dots, I_n), \quad q_n = \varphi_{I_1} \circ \dots \circ \varphi_{I_n}, \quad q_n(x) = S_n x + B_n.$$

Thus q_n has law μ^{*n} . Put $R_n = |S_n|$ and couple

$$X = q_n(X') = B_n + S_n X',$$

where X' is independent of W_n and has law ν . Define the coarse label

$$Q_n = \lfloor B_n/R_n \rfloor, \quad T_n = (S_n, Q_n). \quad (4.1)$$

Knowing T_n determines the signed contraction and locates the translation within one interval of length R_n .

Lemma 4.1. *As $n \rightarrow \infty$,*

$$I(q_n; X) = \alpha \chi n, \quad H(T_n | X) = O(\log n), \quad (4.2)$$

$$H(T_n) = \alpha \chi n + o(n), \quad I(q_n; X | T_n) = o(n), \quad (4.3)$$

$$H(q_n | T_n) = (h_\mu - \alpha \chi)n + o(n). \quad (4.4)$$

In particular, $\alpha \chi \leq h_\mu$.

Proof. By successively conditioning on the known symbols and undoing their composition, the information chain rule gives

$$I(W_n; X) = \sum_{j=1}^n I(I_j; X | I_1, \dots, I_{j-1}) = nI(A; X) = \alpha \chi n.$$

The conditional law of X given W_n depends only on q_n , so $I(q_n; X) = I(W_n; X)$. The bound $I(q_n; X) \leq H(q_n) = nh_\mu + o(n)$ also gives $\alpha \chi \leq h_\mu$.

The scalar product $S_n = \prod_{j=1}^n a_{I_j}$ takes at most $(n+1)^{|\Lambda|}$ values, so $H(S_n) = O(\log n)$. Once S_n is revealed, the integer $Q_n - \lfloor X/R_n \rfloor$ belongs to a fixed finite set, because X' is bounded. Hence $H(T_n | X) = O(\log n)$. Since T_n is a function of q_n , this also gives

$$H(T_n) = I(T_n; X) + H(T_n | X) \leq \alpha \chi n + O(\log n). \quad (4.5)$$

For the matching lower bound, fix $0 < \varepsilon < \chi$ and let $\delta_n = e^{-(\chi - \varepsilon)n}$. The law of large numbers gives

$$\mathbb{P}(R_n \leq \delta_n) \rightarrow 1.$$

On this event, T_n locates X in an interval of length at most a fixed multiple of δ_n . Off this event, compact support bounds the number of possible δ_n -intervals by $e^{O(n)}$. The event itself is determined by T_n . It follows that

$$H(\lfloor X/\delta_n \rfloor | T_n) = o(n).$$

Using (2.1), we conclude that

$$H(T_n) \geq H(\lfloor X/\delta_n \rfloor) - o(n) = \alpha(\chi - \varepsilon)n + o(n).$$

Letting $\varepsilon \downarrow 0$ and applying (4.5) proves $H(T_n) = \alpha\chi n + o(n)$.

The remaining identities now follow from the chain rule:

$$I(q_n; X \mid T_n) = I(q_n; X) - H(T_n) + H(T_n \mid X) = o(n),$$

and, since $H(q_n) = nh_\mu + o(n)$,

$$H(q_n \mid T_n) = H(q_n) - H(T_n) = (h_\mu - \alpha\chi)n + o(n). \quad \square$$

Proof of Theorem 1.1. The upper bound $\alpha \leq \min\{1, h_\mu/\chi\}$ follows from Lemma 4.1. Suppose now that both inequalities are strict. By Proposition 2.2, the law of X satisfies the hypothesis of Proposition 3.1.

Fix $C > \ell_{\max} := \max_i \log(1/r_i)$ and put $s_n = e^{-Cn}$. Condition on a value $T_n = \tau$, and set

$$Y_\tau = B_n/R_n - Q_n \in [0, 1), \quad t_\tau = s_n/R_n, \quad \sigma_\tau = \operatorname{sgn} S_n.$$

Here R_n, Q_n, σ_τ , and t_τ are fixed under the conditioning, and $0 < t_\tau \leq 1$. The variable X' remains independent of Y_τ . Given τ , specifying Y_τ is equivalent to specifying q_n , while subtracting $R_n Q_n$ from X and dividing by R_n gives $Y_\tau + \sigma_\tau X'$. Thus

$$\begin{aligned} I(Y_\tau; Y_\tau + \sigma_\tau X' + t_\tau Z) &\leq I(Y_\tau; Y_\tau + \sigma_\tau X') \\ &= I(q_n; X \mid T_n = \tau). \end{aligned} \quad (4.6)$$

Apply Proposition 3.1 and average over τ . Since $\mathbb{E} \log(1/t_\tau) = (C - \chi)n$, Lemma 4.1 gives, for every $\eta > 0$,

$$\mathbb{E}_\tau H(\lfloor Y_\tau/t_\tau \rfloor) \leq \eta(C - \chi)n + C_\eta(1 + o(n)).$$

First divide by n and let $n \rightarrow \infty$, then let $\eta \downarrow 0$. We obtain

$$\mathbb{E}_\tau H(\lfloor Y_\tau/t_\tau \rfloor) = o(n). \quad (4.7)$$

If all distinct length- n maps having the same signed contraction were separated in translation by at least s_n , then, conditional on $T_n = \tau$, distinct maps would lie in distinct t_τ -intervals. The left side of (4.7) would consequently equal

$$H(q_n \mid T_n) = (h_\mu - \alpha\chi)n + o(n).$$

Its linear coefficient is positive, a contradiction for all sufficiently large n . There are therefore distinct maps with the same signed contraction and translation distance less than e^{-Cn} at every sufficiently large level. Since $C > \ell_{\max}$ was arbitrary, this proves the concentration assertion and (1.4). Exponential separation along infinitely many levels excludes a strict dimension deficit, giving (1.5). $\square$

The fine intervals distinguish maps with their combined word probabilities. Thus exact overlaps are already accounted for by the use of h_μ .

Corollary 4.2. *If all the coefficients a_i, b_i are algebraic, then*

$$\dim \nu = \min \left\{ 1, \frac{h_\mu}{|\chi_\mu|} \right\}.$$

Proof. Choose a number field containing all the coefficients and an integer $D \geq 1$ clearing their denominators. For a word w of length n , $D^n b(\varphi_w)$ is an algebraic integer whose conjugates are at most exponential in n . The algebraic norm therefore gives $|b(\varphi_u) - b(\varphi_v)| \geq e^{-cn}$ whenever the difference is nonzero, for a fixed $c > 0$. Distinct maps with the same signed contraction have different translations. Hence the stronger concentration conclusion in Theorem 1.1 is impossible, proving the formula. $\square$

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